THE TENSOR PRODUCT OF TWO MITTAG-LEFFLER MODULES IS MITTAG-LEFFLER

THE STACKS PROJECT AUTHORS

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Editorial note. The counted proof refers to two same-chapter statements by label (the finiteness of the tensor product of finitely presented modules and the characterisation of Mittag-Leffler modules); both are recorded as navigation stubs pointed at their \label lines, and the arguments consuming them stay inside the counted span.

Editorial note. The counted spans contain no \cite, so the card carries no bibliography.

Allowed standard prerequisites (editorial).

- The characterisation of Mittag-Leffler modules by the existence of compatible maps. Invoked twice in the counted proof (lines 21363-21369 and 21374-21375). Converts the Mittag-Leffler property of the two systems into the existence of the maps whose tensor product is then exhibited.
- Finite presentation of the tensor product of finitely presented modules. Invoked in the counted proof at lines 21359-21361. Shows the approximating modules of the tensor product system are finitely presented.

Source reference. Original lemma 05BS: sources/stacks-algebra/pinned-original-source.tex, source lines 2281–2281. Navigation pointer for \ref{lemma-tensor-finiteness} used inside the counted span (line 21361); the argument that consumes it is retained in the span.

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Source reference. Original lemma 059E: sources/stacks-algebra/pinned-original-source.tex, source lines 21217–21217. Navigation pointer for \ref{proposition-ML-characterization} used inside the counted span (line 21365); the argument that consumes it is retained in the span.

Lemma 0.1. *If R is a ring and M, N are Mittag-Leffler modules over R, then $M \otimes_R N$ is a Mittag-Leffler module.*

Proof. Write $M = \operatorname{colim}_{i \in I} M_i$ and $N = \operatorname{colim}_{j \in J} N_j$ as directed colimits of finitely presented R-modules. Denote $f_{ii'} : M_i \to M_{i'}$ and $g_{jj'} : N_j \to N_{j'}$ the transition maps. Then $M_i \otimes_R N_j$ is a finitely presented R-module (see Lemma 05BS), and $M \otimes_R N = \operatorname{colim}_{(i,j) \in I \times J} M_i \otimes_R M_j$. Pick $(i, j) \in I \times J$. By the definition of a Mittag-Leffler module we have Proposition 059E (3) for both systems. In other words there exist $i' \geq i$ and $j' \geq j$ such that for every choice of $i'' \geq i$ and $j'' \geq j$ there exist maps $a : M_{i''} \to M_{i'}$ and $b : M_{j''} \to M_{j'}$ such that $f_{ii'} = a \circ f_{ii''}$ and $g_{jj'} = b \circ g_{jj''}$. Then it is clear that $a \otimes b : M_{i''} \otimes_R N_{j''} \to M_{i'} \otimes_R N_{j'}$ serves the same purpose for the system $(M_i \otimes_R N_j, f_{ii'} \otimes g_{jj'})$. Thus by the characterization Proposition 059E (3) we conclude that $M \otimes_R N$ is Mittag-Leffler. $\square$